PROJECT-BASEDLEARNING

PHASE-II EVALUATION

College Event Registration Platform

Team ID: DSCPP-III-2026-T188

Department of Computer Science & Engineering

Graphic Era (Deemed to be University), Dehradun

Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID	DSCPP-III-2026-T188
Semester	3rd
Section	AIML1, AIML3, AIML4
Domain	Data Structures and C++

Mentor

Dr. Prabhdeep Singh

Team Members

1. Nikunj Singhal – 2510370169
2. Bhargav Istwal – 2510371598
3. Harshit Kashyap – 2510011536

Phase-II: System Development and Feature Implementation

Project Objectives

1. Develop a functional college event registration system using C++.
2. Implement efficient management of events, students and registrations.
3. Implement event search, registration, cancellation and seat capacity management.
4. Integrate Data Structures and OOP concepts into the project.
5. Test and validate all implemented system modules.

Key Objective

To develop an efficient C++ system for managing college events and registrations.

Proposed Approach

1. Identify students, events, organizers, and registrations.
2. Develop C++ classes and system modules.
3. Manage event registration and seat capacity.
4. Implement searching,sorting and cancellation.
5. Store and display event registration records.

Core Features

- Event Creation and Management
- Student Registration and Cancellation
- Seat Capacity Management
- Event Search and Filtering
- Registration Records and Reports

The Phase-II implementation focuses on converting the proposed architecture into functional event and registration management modules.

01 Event Management

Core Functionality

- Centralized management of all college event details.
- Stores: Event ID, Name, Date, Venue, Category, and Seat Capacity.
- Supports complete operations: Add, Update, Search, Delete, and Display.
- Reduces manual event management and keeps records organized.
- Provides quick access to event information for students and organizers.

Implementation

C ++ Programming
Classes+ Linked Lists

Event Operations
Add → Search → Update → Delete → Display

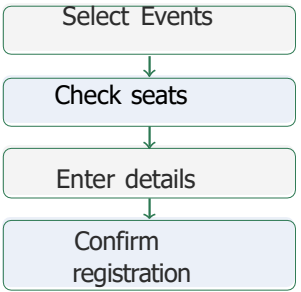
Management Benefit
Fast access to event information

02 Student Registration& Cance lation

Core Functionality

- Allows students to register for selected college events.
- Checks event availability before registration.
- Supports registration cancellation when required.
- Stores student and registration details.
- Updates registration status and available seats.

Registration Flow



Searching Techniquesin C: Linear Search/ Binary Search

03 Event Search and Filtering

Core Functionality

- Allows students to search for events by name or Event ID.
- Helps users find events based on category or date, where supported.
- Displays relevant event details in an organized format.
- Reduces the effort needed to find a suitable event.
- Makes event discovery simpler and more convenient.

Event Record

Event Type
Workshop / Competition

Location
College Campus

Date & Time
Event Schedule

Seat Capacity
Maximum Participants

04 Seat Capacity Management

Core Functionality

- Sets the maximum number of participants for each event.
- Checks seat availability before confirming registration.
- Updates available seats after successful registration.
- Restores seats when a registration is cancelled.
- Prevents registrations beyond the event capacity.
- Displays available seats for easy tracking.

Seat Availability Status



System Architecture / Workflow



Event Management

- Create and update events
- Search event details
- Manage event records

Registration Management

- Student registration
- Registration cancellation
- Registration status

Capacity Management

- Total seat capacity
- Available seats
- Prevent Overbooking

User Input → Processing → Storage → *Registration Confirmation*

DSA & OOP Mapping to System Modules

Concept	Implementation	System Usage
Arrays	C++	Store event records
Linked Lists	C++	Manage events and registration records
Linear search	C++	Search for events
Sorting	C++	Arrange event records
Classes and Objects	C++	Represent events and students
Encapsulation	C++	Organize data and operations
Inheritance	C++	Reusable common functionality
Polymorphism	C++	Support flexible operations

Technology Stack & Development Methodology

Technology Stack

- **Programming:** C++
- **Data Structure:** Arrays and Linked lists
- **Tools / Framework:** STL
- **IDE:** Visual Studio Code
- **Version Control:** Git / GitHub

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

C++ and its Data Structures and OOP concepts help manage college events, student registrations, and seat availability in an organized way.

Progress Achieved

- College event registration requirements identified.
- Student, organizer and event requirements prepared.
- C++ system architecture and workflow prepared.
- Main class and modules identified.
- Event management and registration module developed.
- Seat capacity and registration flow planned.
- Basic data structures selected for record management.

Individual Contribution

Member	Role	Contribution
Nikunj Singhal	Team Lead	Planning and coordination
Bhargav Istwal	Developer & docs	C++ development DSA and documentation
Harshit Kashyap	Testing	Testing and system verification

Roadmap

Phase-I
Proposal & System Design

Phase-II
Feature Development & Testing

Phase-III
Final Implementation & Evaluation

Expected Outcomes

- A functional college event registration system.
- Easy access to event details.
- Simple student registration and cancellation.
- Better tracking of available seats.
- Organized events and registration records.
- Future scope of notification and online registration.

References (cppreference.com)

1. C++ Classes and Objects Documentation
2. C++ Data Structures and Algorithms
3. C++ Standard Library Documentation
4. Git and GitHub Documentation

Project Focus

- Event creation and management
- Student registration and cancellation
- Event search and filtering
- Seat capacity management

Thank You

Questions & Suggestions

College Event Registration Platform